

Hybrid Predictive Guidance with Residual Reinforcement Learning for Autonomous UAS Interception

Technical Report: Proofs of Theoretical Results

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Abstract

This technical report supplements the paper *Hybrid Predictive Guidance with Residual Reinforcement Learning for Autonomous UAS Interception*. It provides detailed proofs of the geometric properties of the proposed Lead-Residual Proximal Policy Optimization (LR-PPO) guidance law. Specifically, the results establish orthogonality and norm bounds for the projected residual, an explicit bound on angular deviation from predictive Lead guidance, exact recovery of the nominal Lead direction for residuals with no orthogonal component, and a positive lower bound on alignment between the executed LR-PPO and Lead commands. These properties constrain the authority of the learned correction but do not imply closed-loop stability or global interception optimality.

Theoretical Properties

Let u_t^{Lead} denote the unit nominal Lead guidance direction. The PPO policy produces a raw residual r_t^{raw} , which is clipped to the unit ball:

$$r_t = \begin{cases} \frac{r_t^{\text{raw}}}{\|r_t^{\text{raw}}\|}, & \|r_t^{\text{raw}}\| > 1, \\ r_t^{\text{raw}}, & \text{otherwise.} \end{cases}$$

Thus,

$$\|r_t\| \leq 1.$$

The component of this residual orthogonal to Lead is

$$r_t^\perp = r_t - (r_t^T u_t^{\text{Lead}}) u_t^{\text{Lead}},$$

and the executed LR-PPO direction is

$$u_t^{\text{LR}} = \text{normalize}(u_t^{\text{Lead}} + \tan(\theta_{\max}) r_t^\perp),$$

where

$$\text{normalize}(x) = \frac{x}{\|x\|}, \quad 0 < \theta_{\max} < \frac{\pi}{2}.$$

The controller evaluated in the accompanying paper uses $\theta_{\max} = 30^\circ$. The results depend only on u_t^{Lead} being a unit nominal direction and therefore also apply when the Lead controller uses its normalized direct-pursuit fallback because no valid analytical intercept solution exists.

Lemma 1. *Let u_t^{Lead} be a unit Lead direction and let the clipped residual satisfy $\|r_t\| \leq 1$. Then*

$$r_t^\perp \perp u_t^{\text{Lead}}, \quad \|r_t^\perp\| \leq 1.$$

Theorem 1. *For*

$$u_t^{\text{LR}} = \text{normalize}(u_t^{\text{Lead}} + \tan(\theta_{\max}) r_t^\perp),$$

with $0 < \theta_{\max} < \pi/2$,

$$\angle(u_t^{\text{LR}}, u_t^{\text{Lead}}) \leq \theta_{\max}.$$

Corollary 1. *If $r_t^\perp = 0$, then*

$$u_t^{\text{LR}} = u_t^{\text{Lead}}.$$

Thus, zero, parallel, or anti-parallel residuals recover the nominal Lead direction exactly.

Corollary 2. *The LR-PPO command always retains positive alignment with Lead:*

$$(u_t^{\text{LR}})^T u_t^{\text{Lead}} \geq \cos(\theta_{\max}) > 0.$$

For $\theta_{\max} = 30^\circ$,

$$\cos(30^\circ) = \frac{\sqrt{3}}{2} \approx 0.866.$$

These results constrain the learned controller geometrically: PPO may modify the predictive Lead command within the prescribed angular neighborhood, but it cannot arbitrarily overwrite or reverse the nominal guidance direction.

Proofs

Proof of Lemma 1

Recall that

$$r_t^\perp = r_t - (r_t^T u_t^{\text{Lead}}) u_t^{\text{Lead}}. \quad (1)$$

Since $\|u_t^{\text{Lead}}\| = 1$, taking the inner product with the Lead direction gives

$$(u_t^{\text{Lead}})^T r_t^\perp = (u_t^{\text{Lead}})^T r_t - (r_t^T u_t^{\text{Lead}}) (u_t^{\text{Lead}})^T u_t^{\text{Lead}} \quad (2)$$

$$= r_t^T u_t^{\text{Lead}} - r_t^T u_t^{\text{Lead}} \quad (3)$$

$$= 0. \quad (4)$$

Therefore,

$$r_t^\perp \perp u_t^{\text{Lead}}. \quad (5)$$

The residual can be decomposed into mutually orthogonal components:

$$r_t = r_t^\perp + (r_t^T u_t^{\text{Lead}}) u_t^{\text{Lead}}. \quad (6)$$

By the Pythagorean relation,

$$\|r_t\|^2 = \|r_t^\perp\|^2 + (r_t^T u_t^{\text{Lead}})^2. \quad (7)$$

Hence,

$$\|r_t^\perp\|^2 \leq \|r_t\|^2. \quad (8)$$

Since the residual is clipped to the unit ball, $\|r_t\| \leq 1$, so

$$\|r_t^\perp\| \leq \|r_t\| \leq 1. \quad (9)$$

This establishes both the orthogonality and norm-bound claims.

Proof of Theorem 1

Define the unnormalized LR-PPO command

$$\tilde{u}_t^{\text{LR}} = u_t^{\text{Lead}} + \tan(\theta_{\max}) r_t^\perp. \quad (10)$$

The executed command is

$$u_t^{\text{LR}} = \frac{\tilde{u}_t^{\text{LR}}}{\|\tilde{u}_t^{\text{LR}}\|}. \quad (11)$$

Let

$$\phi_t = \angle(u_t^{\text{LR}}, u_t^{\text{Lead}}). \quad (12)$$

Normalization changes only magnitude, so ϕ_t is also the angle between $\tilde{u}_t^{\text{LR}}$ and u_t^{Lead} .

Using Lemma 1,

$$(u_t^{\text{Lead}})^T \tilde{u}_t^{\text{LR}} = (u_t^{\text{Lead}})^T u_t^{\text{Lead}} + \tan(\theta_{\max}) (u_t^{\text{Lead}})^T r_t^\perp \quad (13)$$

$$= 1. \quad (14)$$

Thus, $\tilde{u}_t^{\text{LR}} \neq 0$ and

$$0 \leq \phi_t < \frac{\pi}{2}.$$

Equation (10) consists of a unit component along u_t^{Lead} and an orthogonal component of magnitude

$$\tan(\theta_{\max}) \|r_t^\perp\|.$$

Therefore,

$$\tan(\phi_t) = \tan(\theta_{\max}) \|r_t^\perp\|. \quad (15)$$

Lemma 1 gives $\|r_t^\perp\| \leq 1$, hence

$$\tan(\phi_t) \leq \tan(\theta_{\max}).$$

Because tangent is strictly increasing on $[0, \pi/2)$,

$$\phi_t \leq \theta_{\max}.$$

Consequently,

$$\angle(u_t^{\text{LR}}, u_t^{\text{Lead}}) \leq \theta_{\max}. \quad (16)$$

For the controller used in the accompanying paper, $\theta_{\max} = 30^\circ$; therefore the executed LR-PPO direction can deviate by at most 30° from the nominal Lead direction.

Proof of Corollary 1

Suppose

$$r_t^\perp = 0.$$

Then from (10),

$$\tilde{u}_t^{\text{LR}} = u_t^{\text{Lead}}.$$

Because the Lead direction is unit norm,

$$u_t^{\text{LR}} = \frac{u_t^{\text{Lead}}}{\|u_t^{\text{Lead}}\|} = u_t^{\text{Lead}}.$$

In particular, if the clipped residual is parallel to Lead,

$$r_t = \alpha u_t^{\text{Lead}}, \quad |\alpha| \leq 1,$$

then

$$r_t^\perp = \alpha u_t^{\text{Lead}} - \left(\alpha (u_t^{\text{Lead}})^T u_t^{\text{Lead}} \right) u_t^{\text{Lead}} \quad (17)$$

$$= \alpha u_t^{\text{Lead}} - \alpha u_t^{\text{Lead}} \quad (18)$$

$$= 0. \quad (19)$$

Clipping preserves parallel or anti-parallel orientation. Therefore, a raw PPO residual that is zero, parallel, or anti-parallel to the Lead direction also produces $r_t^\perp = 0$, recovering the nominal Lead command exactly.

Proof of Corollary 2

Because u_t^{LR} and u_t^{Lead} are unit vectors,

$$(u_t^{\text{LR}})^T u_t^{\text{Lead}} = \cos(\phi_t), \quad (20)$$

where

$$\phi_t = \angle(u_t^{\text{LR}}, u_t^{\text{Lead}}).$$

By Theorem 1,

$$0 \leq \phi_t \leq \theta_{\max} < \frac{\pi}{2}.$$

Since cosine is decreasing on this interval,

$$\cos(\phi_t) \geq \cos(\theta_{\max}).$$

Therefore,

$$(u_t^{\text{LR}})^T u_t^{\text{Lead}} \geq \cos(\theta_{\max}) > 0. \quad (21)$$

For $\theta_{\max} = 30^\circ$,

$$(u_t^{\text{LR}})^T u_t^{\text{Lead}} \geq \cos(30^\circ) = \frac{\sqrt{3}}{2} \approx 0.866. \quad (22)$$

Hence, LR-PPO always retains a positive directional component along the nominal Lead command and cannot reverse the analytical guidance direction.

Scope of the Results

The preceding results describe geometric properties of the LR-PPO action construction. They guarantee bounded deviation from predictive Lead guidance, exact nominal-controller recovery for residuals without an orthogonal component, and positive alignment with the Lead direction. They do not constitute a proof of closed-loop stability, global interception optimality, or guaranteed interception under arbitrary target motion.